MAST30027 Review Exercises

Detailed worked solutions to Exercises 1–3.

Exercise 1

Poisson regression: R output, likelihood and model comparison

An ecology team records the number of bird calls during a *one-hour* observation at each of $n = 60$ independent sites. The predictors are `elev100` (elevation in units of 100 metres) and `habitat`, with levels `open` (reference), `forest`, and `wetland`. Each site has the *same observation time*, so the models below do *not* include an offset.

Two Poisson regression models have been fitted to these observations:

- **Model A:** `calls ~ elev100` (intercept and elevation).
- **Model B:** `calls ~ elev100 + habitat` (intercept, elevation and habitat).

Model A is *nested within* Model B: Model B has two additional habitat coefficients. We write S for the *saturated model*, which is different from Models A and B. The R output is provided below.

```
> m.A <- glm(calls ~ elev100,
+            family = poisson(link = "log"), data = birds)
> summary(m.A)
Null deviance:     175.628 on 59 degrees of freedom
Residual deviance: 172.242 on 58 degrees of freedom
AIC: 340.001

> m.B <- glm(calls ~ elev100 + habitat,
+            family = poisson(link = "log"), data = birds)
> summary(m.B)
Coefficients:
               Estimate Std. Error z value Pr(>|z|)
(Intercept)     1.31369    0.16260   8.079  6.53e-16
 elev100       -0.06162    0.04930  -1.250  0.2114
 habitatforest  0.37160    0.16281   2.282  0.0225
 habitatwetland -0.05463   0.17765  -0.308  0.7585

Null deviance:     175.628 on 59 degrees of freedom
Residual deviance: 163.846 on 56 degrees of freedom
AIC: 335.606
```

(a) Using **Model B**, interpret the estimated coefficient of `habitatforest` on the original count scale. Calculate the expected number of bird calls for a forest site with `elev100`$= 2$.

(b) For **Models A and B**, let p_A and p_B be their numbers of regression coefficients (including the intercept), and let $\ell_A = \log \mathcal{L}(\hat{\beta}^A)$ and $\ell_B = \log \mathcal{L}(\hat{\beta}^B)$ denote their maximised log-likelihoods. Using the R output and the relations

$$\mathrm{AIC}^A = 2p_A - 2\ell_A, \quad D^A = 2(\ell_S - \ell_A),$$
$$\mathrm{AIC}^B = 2p_B - 2\ell_B, \quad D^B = 2(\ell_S - \ell_B),$$

calculate p_A, p_B, ℓ_A, ℓ_B, and the saturated-model log-likelihood ℓ_S.

(c) Using the residual deviances, compare **Models A and B** with the likelihood-ratio statistic LR^*. State the null hypothesis, test statistic and degrees of freedom. The relevant upper-tail probability is $P(\chi^2_2 \geq 8.396) \approx 0.0150$. Which model is preferred by this test and which by AIC? Explain what assumption underlies the Poisson test.

Detailed worked solution

(a) Forest coefficient and prediction. The Poisson GLM with a log link gives

$$Y_i \mid \mathbf{x}_i \sim \text{Poisson}(\mu_i), \qquad \log \mu_i = \beta_0 + \beta_1 e_i + \beta_2 F_i + \beta_3 W_i,$$

where e_i denotes `elev100`, F_i is the indicator for forest and W_i is the indicator for wetland. The reference category is open habitat ($F_i = W_i = 0$). Thus the complete fitted mean for Model B is

$$\boxed{\hat{\mu}_i = \exp\{1.31369 - 0.06162 e_i + 0.37160 F_i - 0.05463 W_i\}.}$$

The indicator variables are encoded as

Habitat	F_i	W_i
open (reference)	0	0
forest	1	0
wetland	0	1

Thus forest and wetland each have a coefficient relative to the same reference category, open habitat. This is why a factor with three levels has two printed coefficients when an intercept is included.

Interpret `habitatforest`. At a fixed elevation, changing from open habitat ($F = 0, W = 0$) to forest ($F = 1, W = 0$) adds 0.37160 to the log mean. Since the inverse link is exponential, the ratio of fitted mean counts is

$$\frac{\hat{\mu}(\text{forest}, e)}{\hat{\mu}(\text{open}, e)} = \frac{\exp(1.31369 - 0.06162 e + 0.37160)}{\exp(1.31369 - 0.06162 e)} = \exp(0.37160) \approx \boxed{1.450}.$$

Hence, at the same elevation, a forest site has an estimated mean number of calls approximately **45% higher** than an open site. The effect is multiplicative, not an addition of 0.37160 calls.

Predict for a forest site. With `elev100`$= 2$, we have $e = 2$, $F = 1$ and $W = 0$; therefore

$$\hat{\mu} = \exp\{1.31369 - 0.06162(2) + 0.37160\} = \exp(1.56205) \approx \boxed{4.769 \text{ calls}}.$$

The W term is zero for this site. No offset is required because every observation lasts one hour. The fitted mean is the expected number of calls in one hour.

(b) Recovering likelihoods. For these two fitted models, the residual degrees of freedom equal $n - p_A$ and $n - p_B$, respectively. Each parameter count includes the intercept. With $n = 60$,

$$p_B = 60 - 56 = 4, \qquad p_A = 60 - 58 = 2.$$

Model B fits an intercept, an elevation slope and two habitat contrasts; Model A fits an intercept and elevation slope.

For Model A, $\text{AIC}^A = 2p_A - 2\ell_A$, and for Model B, $\text{AIC}^B = 2p_B - 2\ell_B$. Rearranging gives

$$\ell_A = p_A - \frac{\text{AIC}^A}{2}, \qquad \ell_B = p_B - \frac{\text{AIC}^B}{2}.$$

Hence

$$\boxed{\ell_B = p_B - \frac{335.606}{2} = -163.803}, \qquad \boxed{\ell_A = p_A - \frac{340.001}{2} \approx -168.001}.$$

The deviance compares a fitted Poisson model with the saturated model S, which fits the observation-level means. By definition,

$$D^B = 2(\ell_S - \ell_B) \implies \ell_S = \ell_B + \frac{D^B}{2}.$$

Thus

$$\boxed{\ell_S \approx -163.803 + \frac{163.846}{2} = -81.880}.$$

As a useful cross-check, Model A gives $\ell_A + D^A/2 \approx -168.001 + 86.121 \approx -81.880$. The small difference possible in the last decimal place is due to rounding of the displayed R output. Both models use the same observations and therefore share the same saturated model.

(c) Comparing nested models. Model A is nested in Model B. Under

$$H_0 : \beta_{\text{forest}} = \beta_{\text{wetland}} = 0,$$

elevation is retained but habitat provides no additional explanatory effect. Model B has two extra parameters, so the difference in parameter counts is $s = p_B - p_A = 2$.

Why the likelihood-ratio statistic equals a difference of deviances. The likelihood-ratio test compares the best likelihood achievable under Model A (the null hypothesis) with the best likelihood under Model B (the alternative). Its statistic is

$$LR^* = -2\log\left\{\frac{\mathcal{L}(\hat{\beta}^A)}{\mathcal{L}(\hat{\beta}^B)}\right\} = -2(\ell_A - \ell_B) = 2(\ell_B - \ell_A).$$

Here the equality $\log(a/b) = \log a - \log b$ explains the first rearrangement. Since Model A is contained in Model B, maximising over the full parameter space cannot give a smaller log-likelihood: $\ell_B \geq \ell_A$. Hence $LR^* \geq 0$.

Now use the definition of deviance *twice*. For either fitted model M the lecture slides define the *scaled deviance* as

$$\frac{D^M}{\phi} = 2(\ell_S - \ell_M).$$

For an ordinary Poisson model $\phi = 1$, so the scaled deviance and deviance coincide: $D^M = 2(\ell_S - \ell_M)$. Thus

$$\ell_B = \ell_S - \frac{D^B}{2}, \qquad \ell_A = \ell_S - \frac{D^A}{2}.$$

Substitute *both* expressions into the definition of LR^*:

$$\begin{aligned} LR^* &= 2(\ell_B - \ell_A) \\ &= 2\left[\left(\ell_S - \frac{D^B}{2}\right) - \left(\ell_S - \frac{D^A}{2}\right)\right] \\ &= 2\left[-\frac{D^B}{2} + \frac{D^A}{2}\right] \quad \text{(the two } \ell_S \text{ terms cancel)} \\ &= D^A - D^B \\ &= 172.242 - 163.846 = \boxed{8.396}. \end{aligned}$$

The saturated log-likelihood cancels because *the two models fit the same response data under the same Poisson likelihood*. Therefore, we need only the two *residual deviances* printed by R; we do not need a separate printed log-likelihood for each fitted model.

Reference distribution and degrees of freedom. The null hypothesis sets the *two* habitat coefficients to zero. Accordingly,

$$s = p_B - p_A = 4 - 2 = 2 = (n - p_A) - (n - p_B) = 58 - 56.$$

Under H_0, independent Poisson sampling, the assumed Poisson mean–variance relationship and the usual large-sample regularity conditions, Wilks' theorem gives $LR^* \overset{\text{approx.}}{\sim} \chi^2_s = \chi^2_2$. The upper-tail probability is about 0.0150. Since this is below 0.05, *the ordinary Poisson likelihood-ratio test favours Model B*. The algebraic identity between LR^* and the deviance difference does not

require a χ^2 approximation; it is the *test calibration* that depends on the model assumptions and nesting.

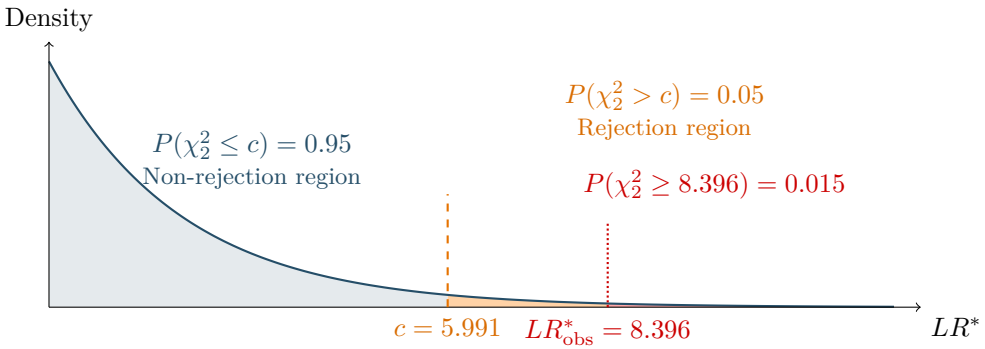
Hypothesis test at significance level $\alpha = 0.05$. We test

$$H_0: \ \beta_{\text{forest}} = \beta_{\text{wetland}} = 0 \qquad \text{against} \qquad H_1: \ \text{at least one of these coefficients is nonzero.}$$

Under H_0, LR^* has an approximate χ^2_2 distribution. The rejection region is in the *right tail*, because a large likelihood-ratio statistic indicates that Model B fits appreciably better than Model A. The $1-\alpha$ critical value is

$$c = \chi^2_{2,0.95} \approx 5.991, \qquad P(\chi^2_2 \le c) = 1-\alpha = 0.95, \quad P(\chi^2_2 > c) = \alpha = 0.05.$$

The following diagram shows the approximate null distribution; the shaded areas represent probabilities *under* H_0.



Here the left area is 0.95 and the entire right-tail area beyond $c = 5.991$ is 0.05. The smaller, darker portion beyond the observed statistic 8.396 is the p-value, approximately 0.0150. Because $LR^*_{\text{obs}} = 8.396 > 5.991$, we reject H_0 at an *alpha* $= 0.05$ level and prefer Model B under the ordinary Poisson assumptions. Note the correct order of operations: $LR^* = -2\log\{\mathcal{L}(\hat{\beta}^A)/\mathcal{L}(\hat{\beta}^B)\}$, *not* the logarithm of a ratio of *log*-likelihoods.

Null versus residual deviance in the supplied output. R also reports a null deviance $D^0 = 175.628$ on 59 degrees of freedom; this compares the *intercept-only* model with S. The residual deviances $D^A = 172.242$ (58 df) and $D^B = 163.846$ (56 df) each compare their specified fitted model with S. To compare A and B, we subtract *their residual deviances*, not their null deviances.

The AIC calculation is a separate comparison: Model B has smaller AIC, $335.606 < 340.001$, so AIC also favours Model B. AIC includes a penalty for the extra fitted parameters; it is not itself a hypothesis test.

Important qualification. Both the usual Poisson likelihood-ratio reference distribution and its standard errors rely on an adequately specified Poisson model, including the working assumption $\text{Var}(Y_i \mid x_i) = \mu_i$ and appropriate independence. Evidence of overdispersion calls the ordinary Poisson inferential conclusion into question. Exercise 2 revisits precisely this issue.

Exercise 2

Overdispersion and quasi-Poisson inference

Continue with the *same observations and Models A and B from Exercise 1.* The quasi-Poisson fits for these two models use exactly the same predictors as their Poisson counterparts. The following additional R-style output is provided.

```
> sum(residuals(m.B, type = "pearson")^2)
[1] 166.272

> m.qA <- glm(calls ~ elev100,
+             family = quasipoisson(link = "log"), data = birds)
> m.qB <- glm(calls ~ elev100 + habitat,
+             family = quasipoisson(link = "log"), data = birds)
> summary(m.qB)
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)     1.31369    0.28018   4.689  1.80e-05
 elev100       -0.06162    0.08495  -0.725    0.4712
 habitatforest  0.37160    0.28054   1.325    0.1907
 habitatwetland -0.05463    0.30611  -0.178    0.8590

(Dispersion parameter for quasipoisson family taken to be 2.969135)
Null deviance:     175.628 on 59 degrees of freedom
Residual deviance: 163.846 on 56 degrees of freedom
AIC: NA

> summary(m.qB)$dispersion
[1] 2.969135

> anova(m.qA, m.qB, test = "F")
Analysis of Deviance Table
Model 1: calls ~ elev100
Model 2: calls ~ elev100 + habitat
  Resid. Df Resid. Dev Df Deviance      F Pr(>F)
1        58    172.242
2        56    163.846  2    8.396 1.4139 0.2517
```

(a) Using $X^2 = \sum_{i=1}^{n} r_P(i)^2$ and the residual degrees of freedom for **Model B**, estimate the dispersion parameter ϕ. Explain what $\hat{\phi}$ indicates about the Poisson variance assumption.

(b) Explain why the fitted coefficients remain unchanged in the quasi-Poisson model while their standard errors change. Verify the reported standard error for `habitatforest`, and calculate a 95% confidence interval for the *forest/open count ratio*, using $t_{56,0.975} = 2.003$. What approximate reference distribution is used for the coefficient test in the quasi-Poisson summary, and how does the conclusion compare with the ordinary Poisson coefficient test from Exercise 1?

(c) Reconstruct the quasi-deviance statistic F_Q comparing **Models A and B** from their deviances and $\hat{\phi}$. State its degrees of freedom and interpret the result at the $\alpha = 0.05$ level. Why does it differ from the Poisson likelihood-ratio test in Exercise 1? Explain briefly why standard AIC and an ordinary likelihood-ratio test are not directly available for quasi-Poisson fits.

Detailed worked solution

(a) Pearson residuals, Pearson's χ^2 statistic and dispersion. In the lectures the variance of an exponential-family response is written

$$E(Y_i \mid x_i) = \mu_i, \qquad \mathrm{Var}(Y_i \mid x_i) = \phi\, v(\mu_i),$$

where $v(\mu)$ is the *variance function*. The *Pearson residual* is defined in class as

$$\boxed{r_P(i) = \frac{y_i - \hat{\mu}_i}{\sqrt{v(\hat{\mu}_i)}}.}$$

It divides the observed-minus-fitted difference by the square root of the *unscaled* fitted variance function. Squaring and summing these residuals gives *Pearson's chi-squared statistic*:

$$\boxed{X^2 = \sum_{i=1}^{n} r_P(i)^2 = \sum_{i=1}^{n} \frac{(y_i - \hat{\mu}_i)^2}{v(\hat{\mu}_i)}.}$$

The R call

```
residuals(m.B, type = "pearson")
```

returns the values $r_P(i)$; summing their squares gives $X^2 = 166.272$.

For Poisson and quasi-Poisson, $v(\mu) = \mu$. Consequently,

$$r_P(i) = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}, \qquad X^2 = \sum_{i=1}^{n} \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i}.$$

As defined in the lecture slides, the dispersion estimator is

$$\boxed{\hat{\phi} = \frac{X^2}{n - p}}$$

where the denominator counts the residual degrees of freedom in the model used to calculate X^2 (including the intercept among its estimated coefficients). We use Model B, for which $p_B = 4$ and $n - p_B = 56$:

$$\boxed{\hat{\phi} = \frac{166.272}{60 - 4} = \frac{166.272}{56} \approx 2.969.}$$

An ordinary Poisson model fixes $\phi = 1$, i.e. $\mathrm{Var}(Y_i \mid x_i) = \mu_i$. A quasi-Poisson model instead allows $\mathrm{Var}(Y_i \mid x_i) = \phi\mu_i$. The estimated value $\hat{\phi} \approx 2.97$ suggests roughly three times the Poisson working conditional variance. It is evidence of overdispersion.

(b) How uncertainty changes. Poisson regression with log link and quasi-Poisson with the same log link use the same mean function and a variance function proportional to μ. The quasi-score equations are the Poisson score equations multiplied by a common factor $1/\phi$, so they have the same solution $\hat{\beta}$ (under the usual fitting conditions). This does *not* mean that uncertainty is the same.

As shown in the GLM lectures, the IWLS-based variance formula

$$\widehat{\mathrm{Var}}(\hat{\beta}) = (X^T \hat{\Sigma}^{-1} X)^{-1}, \qquad \hat{\Sigma}_{ii} = [g'(\hat{\mu}_i)]^2 v(\hat{\mu}_i)\, \hat{\phi}.$$

For the Poisson log link $g(\mu) = \log \mu$, so $g'(\mu) = 1/\mu$ and $v(\mu) = \mu$. Thus $\hat{\Sigma}_{ii} = \hat{\phi}/\hat{\mu}_i$. Multiplying every diagonal entry by the same $\hat{\phi}$ multiplies the estimated coefficient covariance matrix by $\hat{\phi}$:

$$\boxed{\widehat{\mathrm{Var}}_Q(\hat{\beta}) = \hat{\phi}\, \widehat{\mathrm{Var}}_P(\hat{\beta}), \qquad \mathrm{se}_Q(\hat{\beta}_j) = \sqrt{\hat{\phi}}\, \mathrm{se}_P(\hat{\beta}_j).}$$

This is why the numerical coefficient estimates stay fixed even though their estimated standard errors increase. For the forest coefficient, the ordinary Poisson standard error was 0.16281. Consequently,

$$\boxed{\mathrm{se}_Q(\hat{\beta}_{\text{forest}}) = 0.16281\sqrt{2.969} \approx 0.2805.}$$

Why do we use a *t* reference distribution here? For an ordinary Poisson GLM, the dispersion is fixed at $\phi = 1$, and R uses the asymptotic normal (Wald z) reference distribution for individual regression coefficients. In the quasi-Poisson fit, ϕ is *unknown* and is estimated from Pearson's statistic. For a coefficient test under $H_0 : \beta_j = 0$, R therefore reports

$$t^* = \frac{\hat{\beta}_j}{\mathrm{se}_Q(\hat{\beta}_j)} \overset{H_0}{\approx} t_{n-p_B} = t_{56}.$$

The degrees of freedom are $n - p_B = 60 - 4 = 56$ because the dispersion estimate is obtained from the 56 residual degrees of freedom in Model B. For the forest coefficient,

$$t^* = \frac{0.37160}{0.16281\sqrt{2.969135}} \approx \boxed{1.325}, \qquad P(|T_{56}| \geq 1.325) \approx \boxed{0.191}, \quad T_{56} \sim t_{56}.$$

This t reference is an approximation for the quasi-GLM, not an exact finite-sample Student t result. It follows from the estimated scale/dispersion used by R, not from the IWLS algorithm making the observations normally distributed.

The quasi-Poisson `summary(m.qB)` output reports the same coefficient 0.37160, an adjusted standard error of 0.28054, a t-statistic of 1.325 and a two-sided approximate p-value of 0.1907. These values agree with the calculations above (up to rounding).

The estimated *count ratio* remains $\exp(0.37160) \approx 1.450$. For an approximate 95% interval, first work on the coefficient (log-mean) scale using the supplied $t_{56,0.975} = 2.003$ critical value:

$$0.37160 \pm 2.003(0.2805) \approx \boxed{(-0.1903,\ 0.9335)}.$$

Exponentiating both endpoints gives

$$\boxed{\mathrm{CI}_{95\%,Q}(\text{forest/open count ratio}) \approx \left(e^{-0.1903}, e^{0.9335}\right) \approx (0.827,\ 2.543).}$$

The interval includes one, so a zero forest coefficient is compatible with this quasi-Poisson approximation at the $\alpha = 0.05$ level. By contrast, the Poisson output reported $z \approx 2.282$ and $p \approx 0.0225$, suggesting significance under $\phi = 1$. The discrepancy occurs because the quasi-Poisson calculation allows substantially larger uncertainty. The multiplicative effect estimate has *not* changed; the assessed precision has.

(c) The scaled quasi-deviance comparison. The lecture definition of the *quasideviance* is

$$D_Q = -2\sum_{i=1}^{n} \int_{y_i}^{\hat{\mu}_i} \frac{y_i - t}{v(t)}\, dt, \qquad \text{scaled quasideviance} = \frac{D_Q}{\phi}.$$

For the quasi-Poisson variance function $v(t) = t$, this gives the same *unscaled* deviance as ordinary Poisson regression:

$$D_Q = 2\sum_i \left[y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i) \right],$$

with the convention that $0 \log(0/\hat{\mu}_i) = 0$. The fitted means do not change between Poisson and quasi-Poisson, so the displayed *residual deviances* remain $D_Q^A = D^A = 172.242$ and $D_Q^B = D^B = 163.846$. The extra dispersion parameter affects their *scaling*, not the unscaled deviance values.

Our lectures compare nested quasi-models A and B using

$$\boxed{F_Q = \frac{(D_Q^A - D_Q^B)/s}{\hat{\phi}} \overset{\text{approx.}}{\sim} F_{s,n-p_B}, \qquad \hat{\phi} = \frac{X_B^2}{n - p_B}.}$$

Model B has four coefficients, while Model A has two. Thus $s = p_B - p_A = 4 - 2 = 2$ and $n - p_B = 60 - 4 = 56$:

$$F_Q = \frac{(172.242 - 163.846)/2}{166.272/56} = \frac{8.396/2}{2.96914} = \boxed{1.4139}.$$

This is compared with an approximate $F_{2,56}$ distribution under Model A. The printed p-value is 0.2517, which is larger than 0.05, so the quasi-deviance test *does not reject Model A*. The ordinary Poisson test from Exercise 1 instead had $p \approx 0.0150$. The statistics involve the same improvement in fitted means, $D_Q^A - D_Q^B = 8.396$, but evaluate it against *different working variance assumptions*. Under overdispersion, a given deviance improvement is less convincing relative to residual variation.

Recap: mixture models and EM for exercise 3

Connection with the lectures. In class, we used Gaussian components, with separate means μ_k and variances σ_k^2. Exercise 3 uses the *same two-stage mixture construction and the same E-step/M-step principle*, replacing the normal component density by a Poisson probability mass function. Each Poisson component needs only a rate λ_k: its conditional mean and variance both equal λ_k. Hence there is no separate component variance to update.

1. Mixture model and unobserved labels. Following the mixture-model lectures, a mixture observation is generated in two stages: draw an unobserved component Z_i with $\Pr(Z_i = k) = \pi_k$, then draw X_i from the component distribution. For a K-component mixture,

$$f_X(x_i;\theta) = \sum_{k=1}^{K} \pi_k f(x_i;\theta_k), \qquad \pi_k \geq 0, \quad \sum_{k=1}^{K} \pi_k = 1.$$

The observed-data likelihood is a product of *sums*: $L(\theta) = \prod_i \sum_k \pi_k f(x_i;\theta_k)$. In Exercise 3 the components are *Poisson*, not Gaussian: $X_i \mid Z_i = k \sim \text{Poisson}(\lambda_k)$. Do not confuse a mixture distribution with the sum of independent random variables.

2. Observed versus complete data. Write $X = (X_1, \ldots, X_n)$ and $Z = (Z_1, \ldots, Z_n)$. The observed (or incomplete-data) log-likelihood is

$$\ell(\theta) = \log P(X \mid \theta) = \log \sum_{Z} P(X, Z \mid \theta).$$

If the labels were known, the complete-data log-likelihood would be

$$\log P(X, Z \mid \theta) = \sum_{i=1}^{n} \sum_{k=1}^{K} I(Z_i = k)\big[\log \pi_k + \log f(x_i;\theta_k)\big].$$

Here $I(Z_i = k)$ is one for membership in component k and zero otherwise. For a Poisson component, its contribution is $\log \pi_k - \lambda_k + x_i \log \lambda_k - \log(x_i!)$.

3. E-step: posterior membership probabilities. Given current parameter values $\theta^{(t)}$, Bayes' rule gives

$$\boxed{\tau_{ik}^{(t)} = \Pr(Z_i = k \mid X_i = x_i, \theta^{(t)}) = \frac{\pi_k^{(t)} f(x_i;\lambda_k^{(t)})}{\sum_{j=1}^{K} \pi_j^{(t)} f(x_i;\lambda_j^{(t)})}.}$$

The probabilities sum to one over k for each i. They equal the conditional expectations $E[I(Z_i = k) \mid X, \theta^{(t)}]$ and therefore act as *soft-membership weights* in the M-step.

4. M-step: maximise the expected complete log-likelihood. The EM function is

$$\begin{aligned} Q(\theta, \theta^{(t)}) &= E_{Z|X,\theta^{(t)}}[\log P(X, Z \mid \theta)] \\ &= \sum_{i=1}^{n} \sum_{k=1}^{K} \tau_{ik}^{(t)} \big[\log \pi_k - \lambda_k + x_i \log \lambda_k - \log(x_i!)\big]. \end{aligned}$$

Writing $N_k^{(t)} = \sum_i \tau_{ik}^{(t)}$ gives the Poisson-mixture updates

$$\boxed{\pi_k^{(t+1)} = \frac{N_k^{(t)}}{n}, \qquad \lambda_k^{(t+1)} = \frac{\sum_i \tau_{ik}^{(t)} x_i}{N_k^{(t)}}.}$$

The first is the effective proportion in each component; the second is the weighted average count for that component. The rate update is the same kind of weighted mean as the μ_k update in your Gaussian-mixture lectures. Unlike a Gaussian component, a Poisson component has no separate variance parameter to estimate. The E- and M-steps repeat until convergence. EM with exact updates makes the observed log-likelihood nondecreasing, though it can converge to a local rather than global maximum.

Exercise 3

Poisson mixture model and one EM iteration

As in the Gaussian-mixture lectures, each observation is generated by first selecting an unobserved component and then sampling from that component. Here the component distributions are *Poisson*, rather than Gaussian.

A researcher records the number of customer enquiries received in one day at each of three independent shops: $x_1 = 0$, $x_2 = 2$ and $x_3 = 5$. There are two types of shops, but their types are *not observed*. Let $Z_i \in \{1, 2\}$ be the latent shop type and suppose that

$$\Pr(Z_i = k) = \pi_k, \qquad X_i \mid Z_i = k \sim \text{Poisson}(\lambda_k), \qquad \pi_1 + \pi_2 = 1.$$

The unknown parameters are $\theta = (\pi_1, \lambda_1, \lambda_2)$, with $\pi_2 = 1 - \pi_1$. The Poisson probability mass function is

$$f(x; \lambda) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots.$$

For *one iteration* of the EM algorithm, start from

$$\pi_1^{(0)} = \pi_2^{(0)} = 0.5, \qquad \lambda_1^{(0)} = 1, \qquad \lambda_2^{(0)} = 4.$$

(a) Describe the two-stage process used to generate an observation from this mixture. Write $\Pr(X_i = x_i \mid \theta)$ and the *observed-data* log-likelihood $\ell(\theta)$ for these three observations.

(b) **E-step.** Using Bayes' rule, give an expression for $\tau_{i1}^{(0)} = \Pr(Z_i = 1 \mid X_i = x_i, \theta^{(0)})$. Calculate this posterior probability for $x_i = 0, 2, 5$ (to three decimal places). Explain what these probabilities represent.

(c) **M-step.** Put $\tau_{i2}^{(0)} = 1 - \tau_{i1}^{(0)}$ and $N_k^{(0)} = \sum_{i=1}^{3} \tau_{ik}^{(0)}$. Using the E-step weights, calculate the updated mixture proportions $\pi_1^{(1)}$, $\pi_2^{(1)}$ and rates $\lambda_1^{(1)}$, $\lambda_2^{(1)}$. State the update formulas and briefly explain how EM proceeds after this iteration.

Detailed worked solution

(a) Generative process and observed likelihood. For each shop, first choose a latent type Z_i: type 1 with probability π_1 and type 2 with probability $\pi_2 = 1-\pi_1$. Then draw $X_i \mid Z_i = k \sim \text{Poisson}(\lambda_k)$. Since we observe X_i but not Z_i, the law of total probability gives

$$\begin{aligned}\Pr(X_i = x_i \mid \theta) &= \sum_{k=1}^{2} \Pr(X_i = x_i \mid Z_i = k, \theta)\Pr(Z_i = k) \\ &= \pi_1 \frac{e^{-\lambda_1}\lambda_1^{x_i}}{x_i!} + \pi_2 \frac{e^{-\lambda_2}\lambda_2^{x_i}}{x_i!}.\end{aligned}$$

Independence across shops therefore gives the *observed-data* log-likelihood

$$\boxed{\ell(\theta) = \sum_{i=1}^{3} \log\left\{\pi_1 \frac{e^{-\lambda_1}\lambda_1^{x_i}}{x_i!} + \pi_2 \frac{e^{-\lambda_2}\lambda_2^{x_i}}{x_i!}\right\}.}$$

The component sum lies *inside* the logarithm. This prevents the simple component-by-component maximisation we could do if the labels were known and motivates the EM algorithm.

(b) E-step: posterior membership probabilities. By Bayes' rule, the posterior probability is

$$\boxed{\tau_{i1}^{(0)} = \frac{\pi_1^{(0)} f(x_i; \lambda_1^{(0)})}{\pi_1^{(0)} f(x_i; \lambda_1^{(0)}) + \pi_2^{(0)} f(x_i; \lambda_2^{(0)})}.}$$

At the starting values $\pi_1^{(0)} = \pi_2^{(0)} = 1/2$, $\lambda_1^{(0)} = 1$ and $\lambda_2^{(0)} = 4$, the common factor $1/(2x_i!)$ cancels from numerator and denominator:

$$\tau_{i1}^{(0)} = \frac{e^{-1}1^{x_i}}{e^{-1}1^{x_i} + e^{-4}4^{x_i}} = \frac{1}{1 + 4^{x_i}e^{-3}}.$$

Substituting the three observed counts,

$$\begin{aligned}\tau_{11}^{(0)} &= \frac{1}{1+e^{-3}} \approx 0.952574, \\ \tau_{21}^{(0)} &= \frac{1}{1+16e^{-3}} \approx 0.556609, \\ \tau_{31}^{(0)} &= \frac{1}{1+1024e^{-3}} \approx 0.019237.\end{aligned}$$

The complementary responsibilities are $\tau_{i2}^{(0)} = 1 - \tau_{i1}^{(0)}$:

x_i	$\tau_{i1}^{(0)}$	$\tau_{i2}^{(0)}$
0	0.952574	0.047426
2	0.556609	0.443391
5	0.019237	0.980763

The observation $x_i = 0$ is very likely to come from the low-rate component, while $x_i = 5$ is very likely to come from the high-rate component. But the actual Z_i remains unobserved: the E-step assigns probabilities, not hard labels.

(c) M-step: update mixture weights and Poisson rates. The expected complete log-likelihood, with current responsibilities held fixed, is

$$Q(\theta, \theta^{(0)}) = \sum_{i=1}^{3}\sum_{k=1}^{2} \tau_{ik}^{(0)}\left[\log \pi_k - \lambda_k + x_i \log \lambda_k - \log(x_i!)\right].$$

Let $N_k^{(0)} = \sum_{i=1}^3 \tau_{ik}^{(0)}$. Their values are

$$N_1^{(0)} = 0.952574 + 0.556609 + 0.019237 \approx 1.528421,$$
$$N_2^{(0)} = 3 - N_1^{(0)} \approx 1.471579.$$

Maximising Q yields the formulas

$$\boxed{\pi_k^{(1)} = \frac{N_k^{(0)}}{3}, \qquad \lambda_k^{(1)} = \frac{\sum_{i=1}^3 \tau_{ik}^{(0)} x_i}{N_k^{(0)}}.}$$

Hence the updated mixture weights are

$$\boxed{\pi_1^{(1)} \approx 0.509474, \qquad \pi_2^{(1)} \approx 0.490526.}$$

The component rates become

$$\lambda_1^{(1)} = \frac{0(0.952574) + 2(0.556609) + 5(0.019237)}{1.528421} \approx \boxed{0.791278},$$
$$\lambda_2^{(1)} = \frac{0(0.047426) + 2(0.443391) + 5(0.980763)}{1.471579} \approx \boxed{3.934952}.$$

These are *weighted means of the observed counts*, not sample means based on assigning each shop definitively to a component. This is directly analogous to updating the Gaussian component means in class. Here there is no update for σ_k^2 because the Poisson component variance is already determined by its rate λ_k.

Derivation of the M-step formulas. For a fixed component k, terms involving λ_k in Q are $\sum_i \tau_{ik}^{(0)}[-\lambda_k + x_i \log \lambda_k]$. Differentiate and set the derivative to zero:

$$\frac{\partial Q}{\partial \lambda_k} = \sum_i \tau_{ik}^{(0)} \left(-1 + \frac{x_i}{\lambda_k}\right) = 0 \quad \Longrightarrow \quad \lambda_k^{(1)} = \frac{\sum_i \tau_{ik}^{(0)} x_i}{\sum_i \tau_{ik}^{(0)}}.$$

The second derivative is $-\sum_i \tau_{ik}^{(0)} x_i / \lambda_k^2 \leq 0$, so this stationary point maximises the relevant terms (for strictly positive weighted counts). For the mixture weights, maximise $N_1^{(0)} \log \pi_1 + N_2^{(0)} \log(1 - \pi_1)$:

$$\frac{N_1^{(0)}}{\pi_1} - \frac{N_2^{(0)}}{1 - \pi_1} = 0 \quad \Longrightarrow \quad \pi_1^{(1)} = \frac{N_1^{(0)}}{N_1^{(0)} + N_2^{(0)}} = \frac{N_1^{(0)}}{3}.$$

Continuing EM. Use $\theta^{(1)}$ in Bayes' rule for another E-step, then maximise the new expected complete log-likelihood in the next M-step. Repeat until the observed-data log-likelihood changes by less than a specified tolerance (or a maximum number of iterations is reached). Exact EM updates cannot decrease this log-likelihood, but multiple starting values are helpful because convergence to the global maximum is not guaranteed.

Please remember! The E-step computes posterior probabilities of unobserved group membership; the M-step updates the *rates* and mixture proportions using those probabilities as weights. This is the same EM principle as in the Gaussian-mixture lectures, despite the different component distribution.

Connection to Exercise 2. If the two groups have different Poisson rates, the combined population can be overdispersed even though each component has variance equal to its mean. This observation is for interpretation only; it is not required for the EM calculations.